

Introduction To Flight Mechanics

Tutorial 4

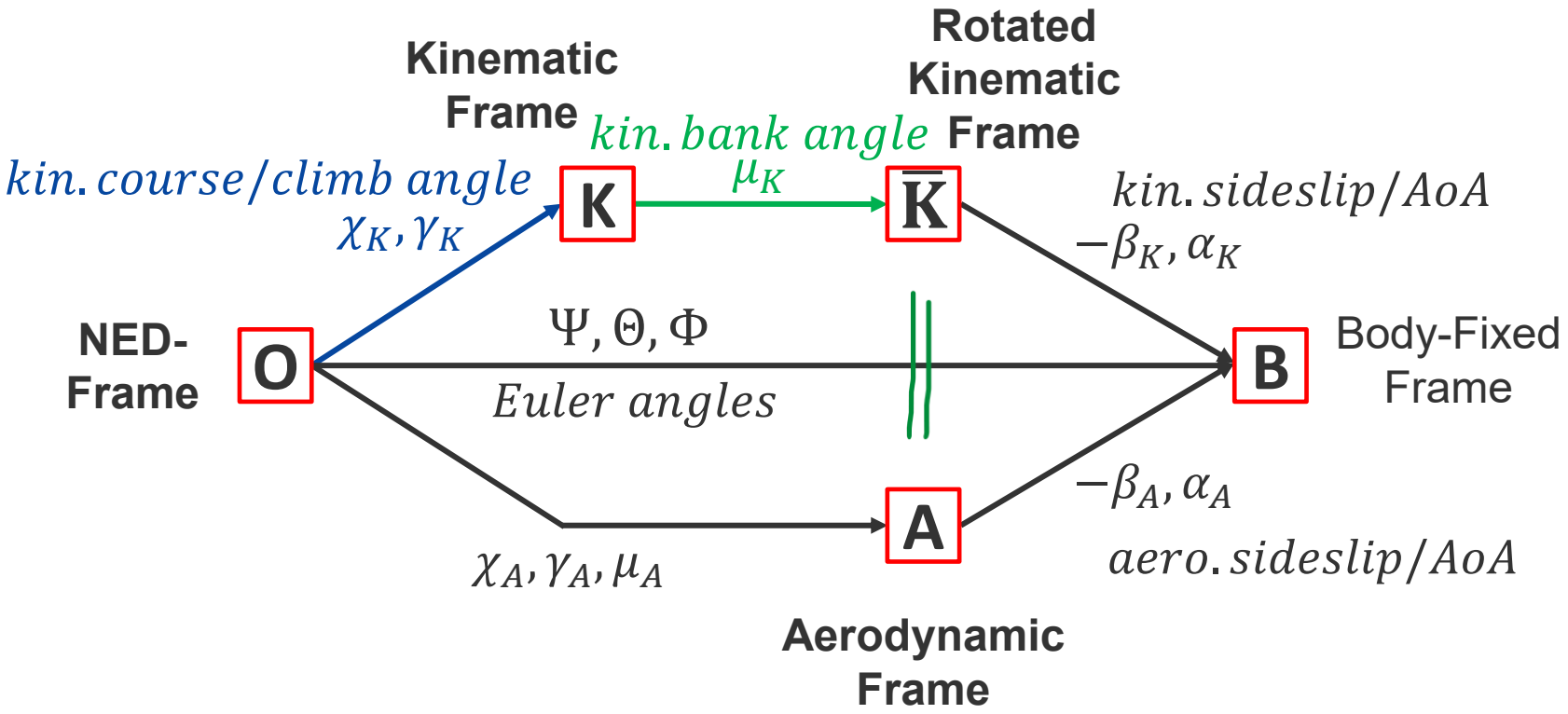
Exercise 1: Pointmass-Equations and Forces

A 1-engine propeller aircraft (piston engine) is flying in a windfree standard ISA-atmosphere without a bank angle in a straight flight with a climb angle γ_K of 0.5°. You are given the following data.

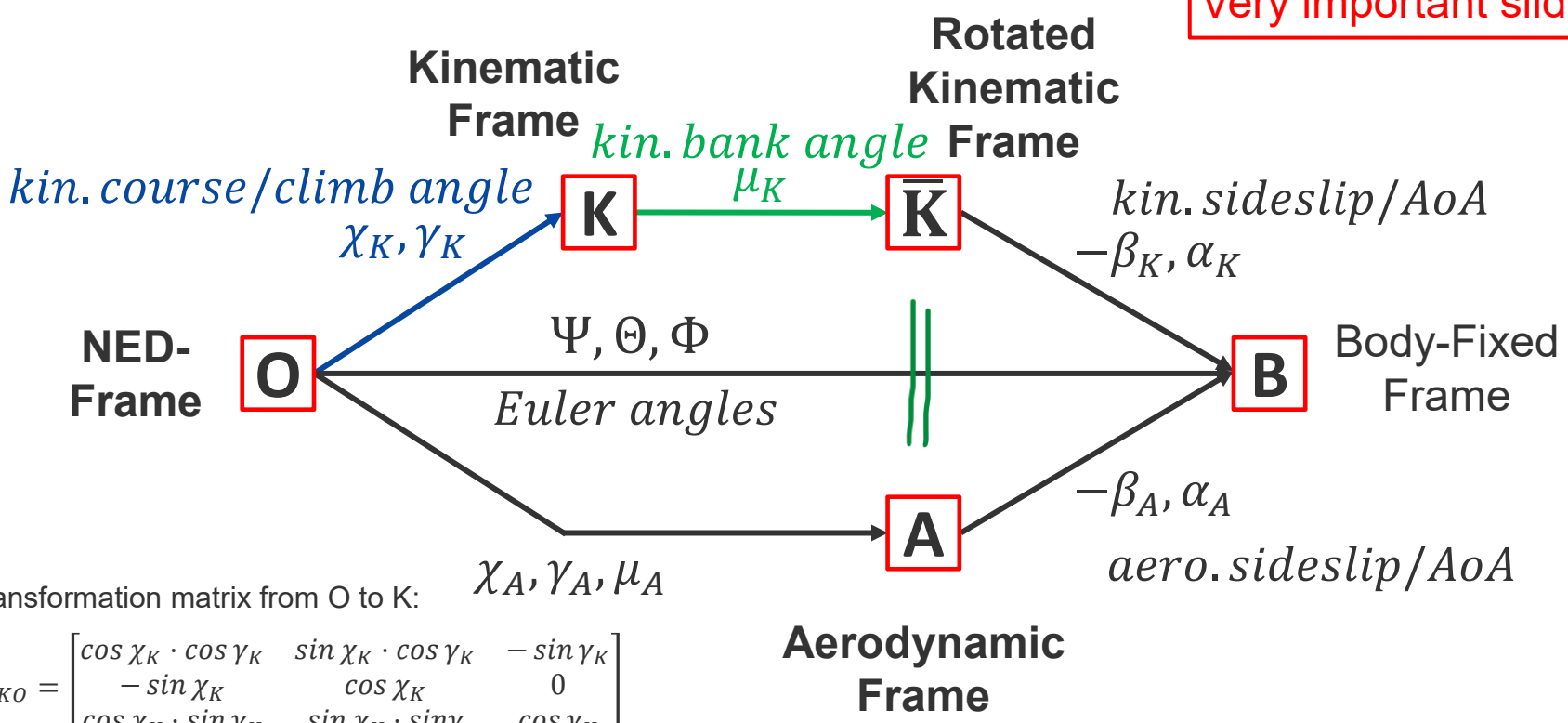
In the following you can assume that no aerodynamic side-forces act on the aircraft and that the kinematic bank angle μ_K equals the euler bank angle ϕ.

a) What is the required GPS velocity such that the z-component of the total force in the kinematic frame is zero?
b) Calculate the total force acting on the aircraft. State the result in a suitable coordinate frame.
c) Calculate the current time derivative of the aircraft states.
d) What happens to the time derivatives if the aircraft flies with a kinematic bank angle $\mu_K = 30°$ if all other states remain the same?

Very important slide!



Very important slide!

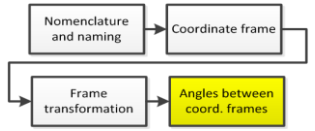


Transformation matrix from O to K:

$$M_{KO} = \begin{bmatrix} \cos\chi_K \cdot \cos\gamma_K & \sin\chi_K \cdot \cos\gamma_K & -\sin\gamma_K \\ -\sin\chi_K & \cos\chi_K & 0 \\ \cos\chi_K \cdot \sin\gamma_K & \sin\chi_K \cdot \sin\gamma & \cos\gamma_K \end{bmatrix}$$

Note:

1) you **do not need** to learn those transformation matrices by heart!
2) you **do need** to know the order of rotation though!
3) you **do need** to know how to perform single-axis rotations w.r.t. some specific angle!



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2) you **do need** to know the order of rotation though!
3) you do need to know how to perform single-axis rotations w.r.t. some specific angle!

Rotation around x-Axis

$$\boldsymbol{M}_{AB} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha \\ 0 & -\sin\alpha & \cos\alpha \end{pmatrix}$$

Rotation around y-Axis

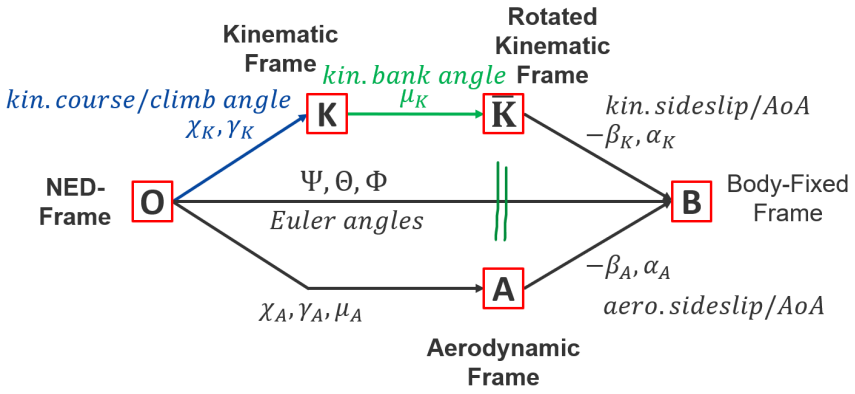
$$\boldsymbol{M}_{AB} = \begin{pmatrix} \cos\alpha & 0 & -\sin\alpha \\ 0 & 1 & 0 \\ \sin\alpha & 0 & \cos\alpha \end{pmatrix}$$

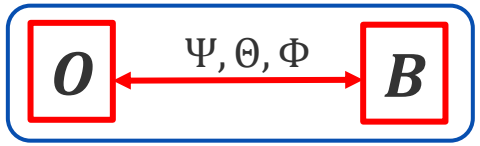
Rotation around z-Axis

$$\boldsymbol{M}_{AB} = \begin{pmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Successive rotations around coordinate system axis are <u>not</u> commutative!

Rotation ORDER MATTERS





→ **Sequence of rotation Ψ, Θ, Φ**

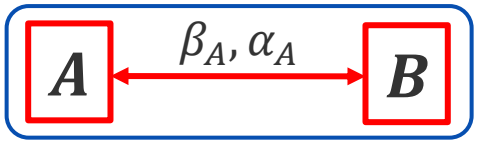
Rotation 3:
Bank angle Φ
Rotation axis: x_B

Rotation 2:
Pitch angle Θ
Rotation axis: k_2

Rotation 1:
Azimuth angle Ψ
Rotation axis: z_0

$$\boldsymbol{M}_{BO} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\Phi & \sin\Phi \\ 0 & -\sin\Phi & \cos\Phi \end{bmatrix} \begin{bmatrix} \cos\Theta & 0 & -\sin\Theta \\ 0 & 1 & 0 \\ \sin\Theta & 0 & \cos\Theta \end{bmatrix} \begin{bmatrix} \cos\Psi & \sin\Psi & 0 \\ -\sin\Psi & \cos\Psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

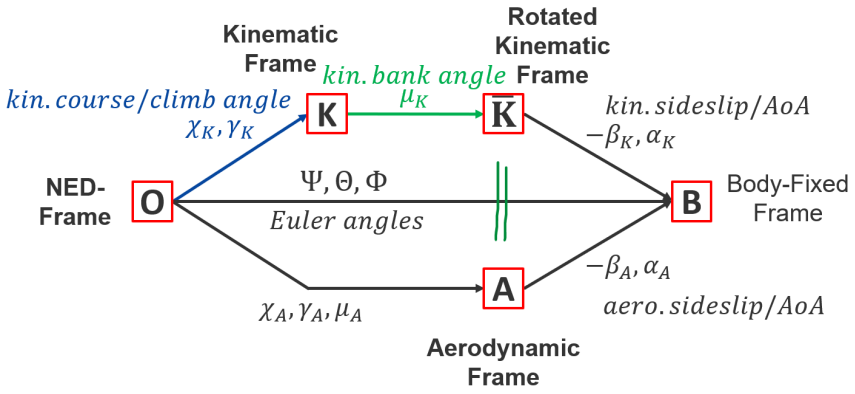
$$\boldsymbol{M}_{BO} = \begin{pmatrix} \cos\Psi \cdot \cos\Theta & \sin\Psi \cdot \cos\Theta & -\sin\Theta \\ \cos\Psi \cdot \sin\Theta \cdot \sin\Phi - \sin\Psi \cdot \cos\Phi & \sin\Psi \cdot \sin\Theta \cdot \sin\Phi + \cos\Psi \cdot \cos\Phi & \cos\Theta \cdot \sin\Phi \\ \cos\Psi \cdot \sin\Theta \cdot \cos\Phi + \sin\Psi \cdot \sin\Phi & \sin\Psi \cdot \sin\Theta \cdot \cos\Phi - \cos\Psi \cdot \sin\Phi & \cos\Theta \cdot \cos\Phi \end{pmatrix}$$

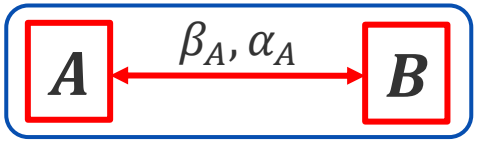




→ **From A to B - Sequence of rotation ?**

→ ***How would you calculate M_{BA} ?***





sequence of rotation $-\beta_A, \alpha_A$

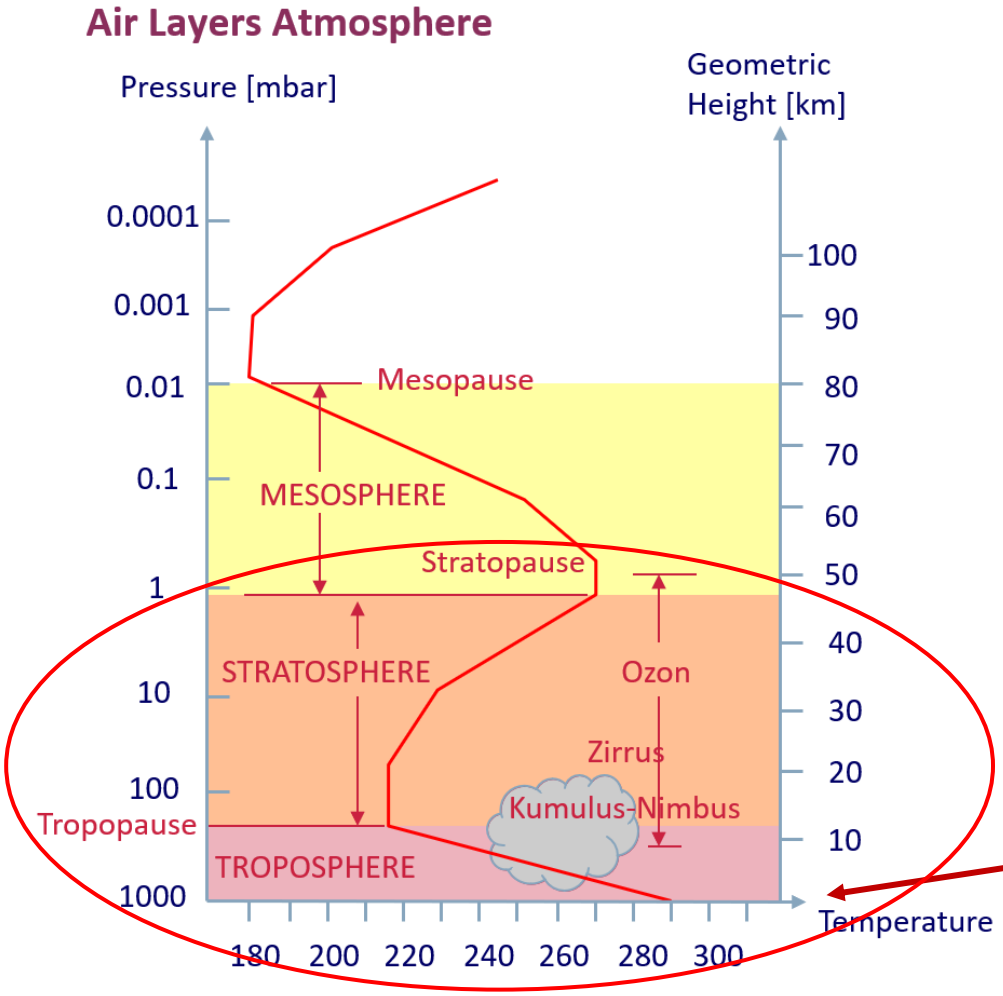
Rotation 2:
Angle of Attack α_A,
rotation axis: y_B

Rotation 1:
neg. sideslip angle $-\beta_A$,
rotation axis: z_A

$$\boldsymbol{M}_{BA} = \begin{bmatrix} \cos\alpha_A & 0 & -\sin\alpha_A \\ 0 & 1 & 0 \\ \sin\alpha_A & 0 & \cos\alpha_A \end{bmatrix} \begin{bmatrix} \cos -\beta_A & \sin -\beta_A & 0 \\ -\sin -\beta_A & \cos -\beta_A & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\boldsymbol{M}_{BA} = \begin{pmatrix} \cos\alpha_A \cos\beta_A & -\cos\alpha_A \sin\beta_A & -\sin\alpha_A \\ \sin\beta_A & \cos\beta_A & 0 \\ \sin\alpha_A \cos\beta_A & -\sin\alpha_A \sin\beta_A & \cos\alpha_A \end{pmatrix}$$

Standard Atmosphere– ISO 2533



Geometric Height h
Height over WGS-84 reference ellipsoid ("height over sea level")

=

Geodetic Height h_{geod}

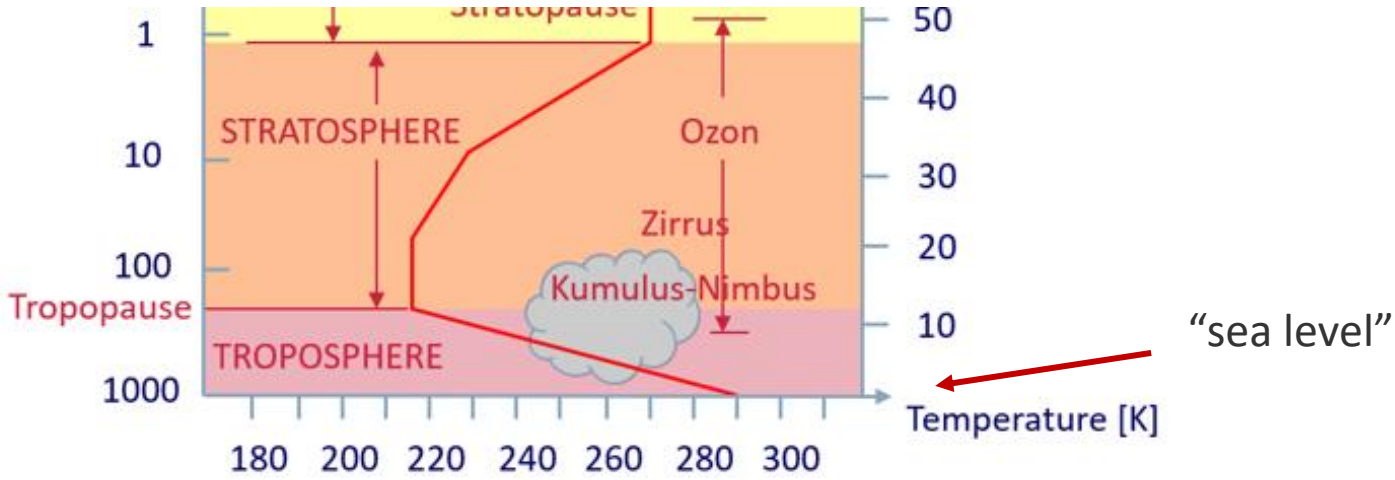
=

GPS measured Height h_{GPS}

Vertical distance above the WGS-84 ellipsoid. This "reference ellipsoid" serves as an idealized, mathematical model of the Earth's shape, rather than the physical Earth's surface.

"sea level"

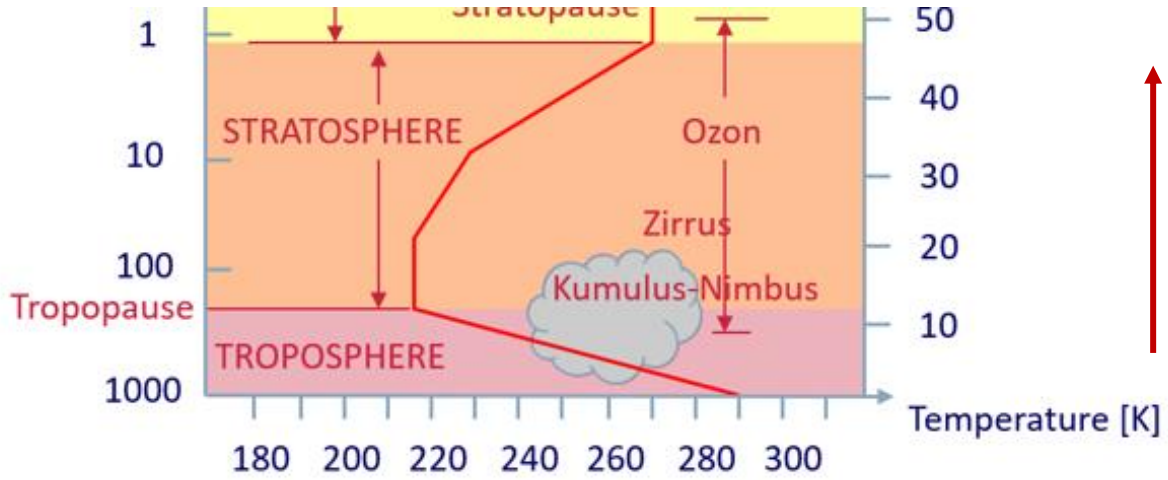
Key Facts ISA



"sea level"

Standard-values for pressure, temperature and density on mean sea level (MSL)

$$p_{S,MSL} = 1{,}01325 \cdot 10^5 \frac{N}{m^2} \qquad T_{S,MSL} = 288{,}15\ K \qquad \rho_{S,MSL} = 1{,}2250 \frac{kg}{m^3}$$



decrease of earth acceleration $g(h)$

Geopotential height H_g

reflects $g(h)$

$$H_G = \frac{r_E \cdot h}{r_E + h}$$

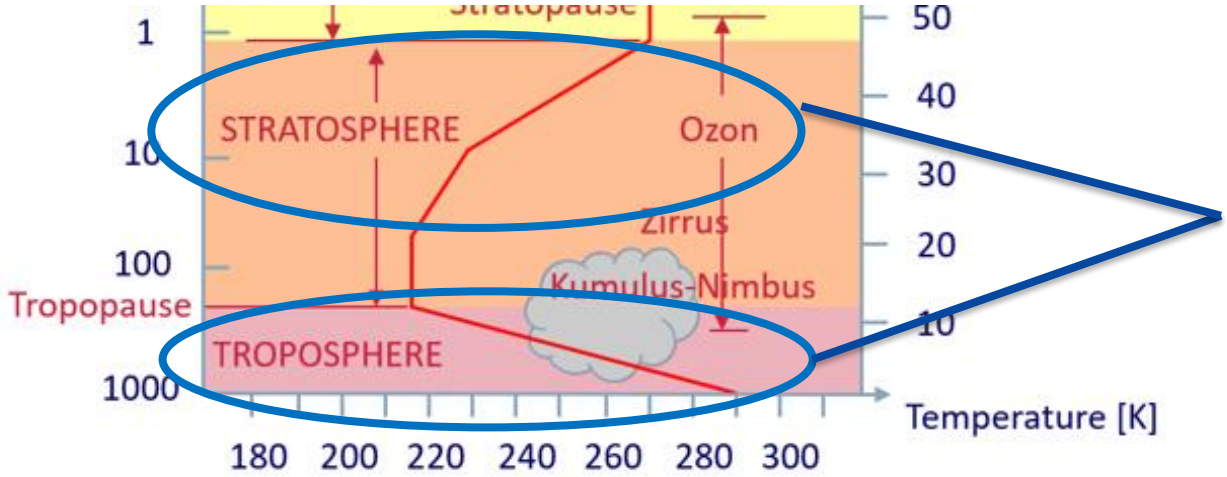
Earth radius according to ISO 2533: $r_E = 6356766m$

Geodetic height: $h = h_{GPS}$

Using the geopotential height within the (following) equations, we can assume the **earth acceleration** to be **constant**:

$$g_s = 9.80665\ \mathrm{m/s^2}$$

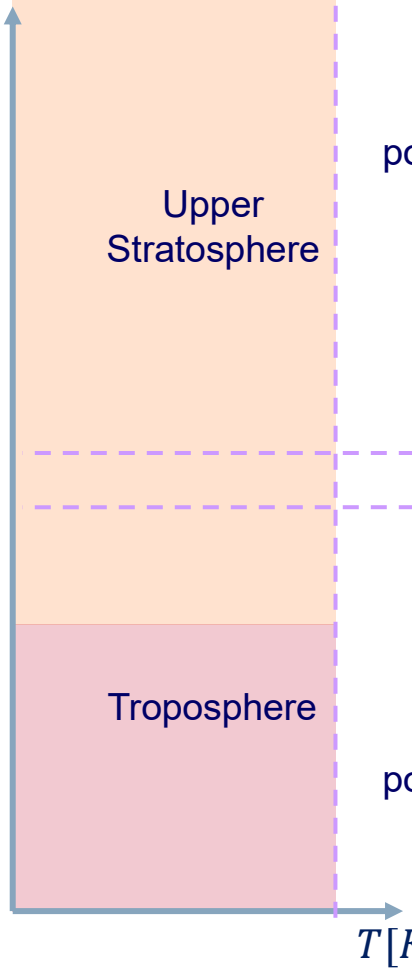
Key Facts ISA



Derivations assumed linear temperature curve

Holds for Troposphere/ Upper stratosphere:

$$H_G = \frac{r_E \cdot h}{r_E + h}$$



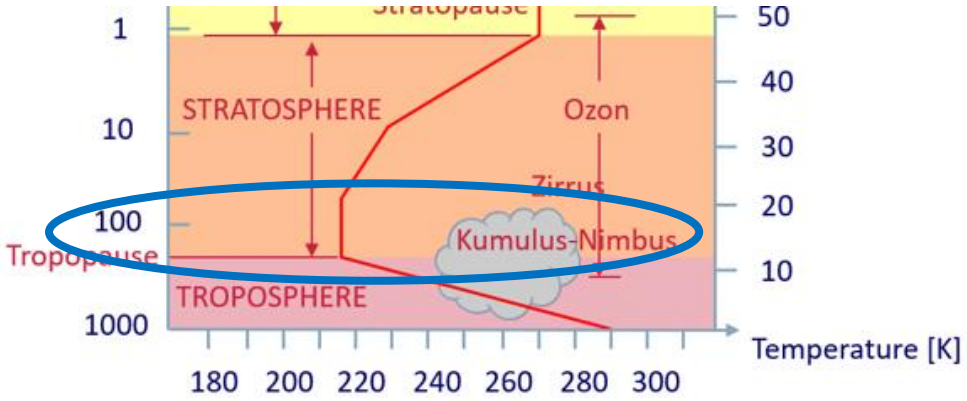
Layer	Features Of Layer	Atmosphere Equations	Data	Height
Upper Stratosphere	polytrope	$\frac{T(h)}{T_{20}} = \left[1 + \frac{\gamma_{str}}{T_{20}}(H_G(h) - H_{G20})\right]$ $\frac{p(h)}{p_{20}} = \left[1 + \frac{\gamma_{str}}{T_{20}}(H_G(h) - H_{G20})\right]^{\frac{n_{str}}{n_{str}-1}}$ $\frac{\rho(h)}{\rho_{20}} = \left[1 + \frac{\gamma_{str}}{T_{20}}(H_G(h) - H_{G20})\right]^{\frac{1}{n_{str}-1}}$	$T_{20} = 216{,}65K$ $\rho_{20} = 0{,}088\frac{kg}{m^3}$ $p_{20} = 547{,}488Pa$ $H_{G20} = 20000m$ $n_{Str} = 0{,}972$ $\gamma_{Str} = +1{,}0e-3$	20 to 32 km
Troposphere	polytrope	$\frac{T(h)}{T_{S,MSL}} = \left[1 + \frac{\gamma_{Tr}}{T_{S,MSL}}H_G(h)\right]$ $\frac{p(h)}{p_{S,MSL}} = \left[1 + \frac{\gamma_{Tr}}{T_{S,MSL}}H_G(h)\right]^{\frac{n_{Tr}}{n_{Tr}-1}}$ $\frac{\rho(h)}{\rho_{S,MSL}} = \left[1 + \frac{\gamma_{Tr}}{T_{S,MSL}}H_G(h)\right]^{\frac{1}{n_{Tr}-1}}$	$T_{S,MSL} = 288{,}15K$ $\rho_{S,MSL} = 1{,}225\frac{kg}{m^3}$ $p_{S,MSL} = 101325Pa$ $H_{GS,MSL} = 0m$ $n_{Tr} = 1{,}235$ $\gamma_{Tr} = -6{,}5e-3$	-2 to 11 km

$g_S = 9{,}80665\frac{m}{s^2}$ $R = 287{,}05\frac{J}{kg\,K}$ $n = \frac{g_S}{g_S + \gamma R}$ $\gamma = -\left(\frac{n-1}{n}\right)\frac{g_S}{R}$

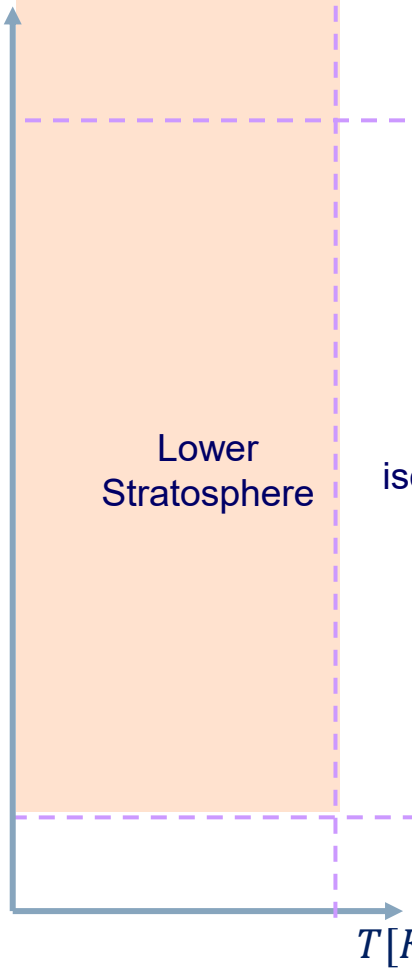
Key Facts ISA

Derivations assumed constant Temperature

lower stratosphere:



$$H_G = \frac{r_E \cdot h}{r_E + h}$$



Layer	Features Of Layer	Atmosphere Equations	Data	Height
Lower Stratosphere	isotherm	$\frac{T(h)}{T_{11}} = 1$ $\frac{p(h)}{p_{11}} = e^{-\frac{g_S}{R\,T_{11}}(H_G(h)-H_{G11})}$ $\frac{\rho(h)}{\rho_{11}} = e^{-\frac{g_S}{R\,T_{11}}(H_G(h)-H_{G11})}$	$T_{11} = 216{,}65K$ $\rho_{11} = 0{,}364\frac{kg}{m^3}$ $p_{11} = 22632Pa$ $H_{G11} = 20000m$ $n_{Str} = 1$ $\gamma_{Str} = 0$	11 to 20 km

$g_S = 9{,}80665\frac{m}{s^2}$ $R = 287{,}05\frac{J}{kg\,K}$

Standard Atmosphere– ISO 2533

Note: you need to know these quantities in the exam!

MSL density ISA standard atmosphere:	$\rho_{s,MSL} = 1.225 \frac{kg}{m^3}$
MSL pressure ISA standard atmosphere:	$p_{s,MSL} = 1.01325e5\ Pa$
MSL temperature ISA standard atmosphere:	$T_{s,MSL} = 288.15\ K$
Gravitational acceleration:	$g_S = 9.80665 \frac{m}{s^2}$
Speed of sound MSL:	$a_{s,MSL} = 340 \frac{m}{s}$
Specific gas constant:	$R = 287.05 \frac{J}{kg\ K}$
Polytropic exponent (Troposphere):	$n = 1.235$
Isentropic exponent	$\kappa = 1{,}4$
Earth radius according to ISO 2533:	$r_E = 6356766\ m$

Key Facts

A 1-engine propeller aircraft (**piston engine**) flies in **windfree standard atmosphere without a bank angle** in a straight flight with a climb angle γ_K **of 0.5°.**

Geodetic Longitude	λ	11.669284 deg
Geodetic Latitude	μ	48.265906 deg
GPS-height	h	5000 m
Climb angle	γ_K	0,5°
Reference thrust MSL	T_i	5000 N
Reference density	ρ_i	$\rho_{s,MSL}$
Reference velocity	V_{ref}	80 kts
Density coefficient	n_ρ	0.75
Lift coefficient	C_L	1
Drag coefficient	$C_{D,0}$	0,012
K - factor	k	0,016
Reference area	S_{ref}	20 m^2
Aircraft mass	m	2000 kg

Assumptions:

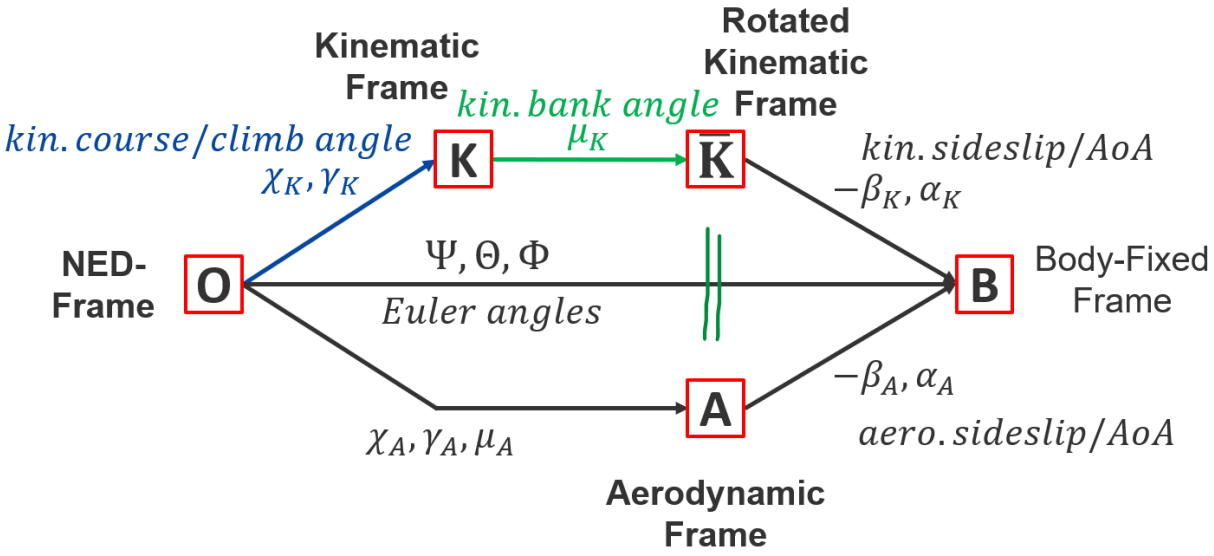
I. No aerodynamic cross forces (Q)

II. Kinematic bank μ_K and euler bank angle ϕ are the same

Key Facts

A 1-engine propeller aircraft (**piston engine**) flies in **windfree standard atmosphere without a bank angle** in a straight flight with a climb angle γ_K **of 0.5°.**

(1) windfree: $A = \bar{K}$
(2) $\Phi = 0$
(3) $\gamma_K = 0.5\,°$



Assumptions:

I. No aerodynamic cross forces (Q)

II. Kinematic bank μ_K and euler bank angle ϕ are the same

Exercise 1: Pointmass-Equations and Forces

A 1-engine propeller aircraft (piston engine) is flying in a windfree standard ISA-atmosphere without a bank angle in a straight flight with a climb angle γ_K of 0.5°. You are given the following data.

In the following you can assume that no aerodynamic side-forces act on the aircraft and that the kinematic bank angle μ_K equals the euler bank angle ϕ.

a) What is the required GPS velocity such that the z-component of the total force in the kinematic frame is zero?

b) Calculate the total force acting on the aircraft. State the result in a suitable coordinate frame.

c) Calculate the current time derivative of the aircraft states.

d) What happens to the time derivatives if the aircraft flies with a kinematic bank angle $\mu_K = 30°$ if all other states remain the same?

Force equilibrium z_K

a) What is the **required GPS velocity** such that **the z-component of the total force** in the **kinematic frame** is **zero**?

$$\left(F_{T,z}\right)_K = 0 \quad \rightarrow \quad V = ?$$

a) What is the **required GPS velocity** such that **the z-component of the total force** in the **kinematic frame** is zero?

$$\left(\vec{F_T}\right)_K = \left(\vec{F}_A^A\right)_K + \left(\vec{F}_P^G\right)_K + \left(\vec{F}_G^G\right)_K$$

with:

Try to fill the gaps



$$\left(\vec{F}_A^A\right)_A = \left(\quad\right)_A$$

$$\left(\vec{F}_P^G\right)_K = \left(\quad\right)_K \qquad \left(\vec{F}_G^G\right)_O = \left(\quad\right)_O$$

We can compute these rather easily!

Force equilibrium z_K

a) What is the **required GPS velocity** such that **the z-component of the total force** in the **kinematic frame** is zero?

$$\left(\vec{F_T}\right)_K = \left(\vec{F}_A^A\right)_K + \left(\vec{F}_P^G\right)_K + \left(\vec{F}_G^G\right)_K$$

with:

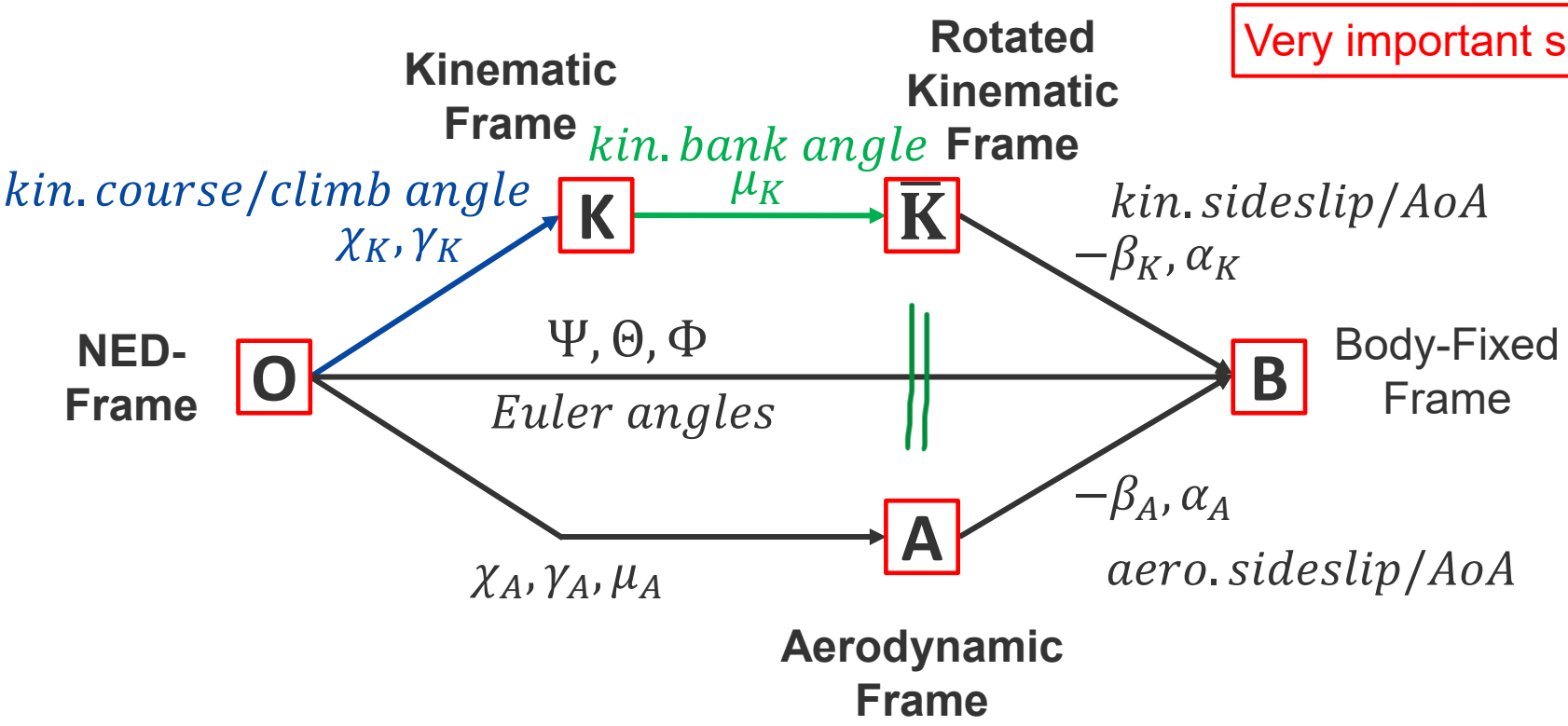
$$\left(\vec{F}_A^A\right)_A = \begin{bmatrix} X_A^A \\ Y_A^A \\ Z_A^A \end{bmatrix}_A = \begin{bmatrix} -D \\ Q \\ -L \end{bmatrix}_A = \bar{q}\, S_{ref} \begin{bmatrix} -C_D \\ C_Q \\ -C_L \end{bmatrix}_A$$

$$\left(\vec{F}_P^G\right)_K = \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K \qquad \left(\vec{F}_G^G\right)_O = m\, g_S \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_O$$

We can compute these rather easily!

We need to transform $\left(\vec{F}_G^G\right)_O$ and $\left(\vec{F}_A^A\right)_A$ into the K-frame!

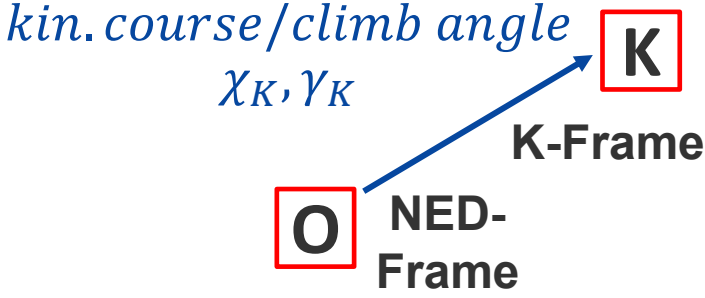
Very important slide!



$\left(\vec{F}_G^G\right)_O$ 1) For the **gravitational force component:**
O frame to the K frame!

$\left(\vec{F}_A^A\right)_A$ 2) For the **aerodynamic force component:**
A frame to the K frame!

((1) no wind -> A = $\overline{K}$ frame)



1. Transform the **gravitational force** from O frame to K frame:

$$\left(\vec{F}_G^G\right)_K = M_{KO} \cdot \left(\vec{F}_G^G\right)_O$$

$$\left(\vec{F}_G^G\right)_K = \begin{bmatrix} \cos\chi_K \cdot \cos\gamma_K & \sin\chi_K \cdot \cos\gamma_K & -\sin\gamma_K \\ -\sin\chi_K & \cos\chi_K & 0 \\ \cos\chi_K \cdot \sin\gamma_K & \sin\chi_K \cdot \sin\gamma & \cos\gamma_K \end{bmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ m\,g \end{pmatrix}_O = \begin{pmatrix} -\sin\gamma_K\, m\, g \\ 0 \\ \cos\gamma_K\ m\, g \end{pmatrix}_K$$

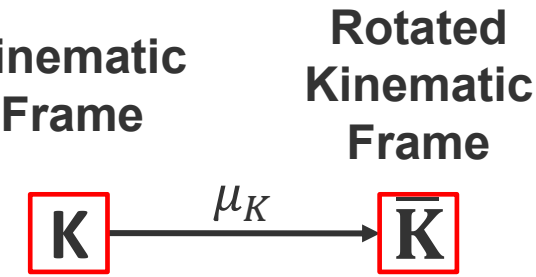
Note:
1) you **do not need** to learn those transformation matrices by heart!
2) you **do need** to know the order of rotation though!
3) you **do need** to know how to perform single-axis rotations w.r.t. some specific angle!

Sequence of rotation χ_K, γ_K

Rotation 2: Kin. climb angle γ_K, Rotation axis: y_K
Rotation 1: Kin. azimuth angle χ_K, Rotation axis: z_O

$$M_{KO} = \begin{bmatrix} \cos\gamma_K & 0 & -\sin\gamma_K \\ 0 & 1 & 0 \\ \sin\gamma_K & 0 & \cos\gamma_K \end{bmatrix} \begin{bmatrix} \cos\chi_K & \sin\chi_K & 0 \\ -\sin\chi_K & \cos\chi_K & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Transform **aerodynamic force** from A frame to K frame:



No wind:

$$\underbrace{\left(\vec{F}_A^G\right)_A = \left(\vec{F}_A^G\right)_{\bar{K}}}_{\textbf{Assumption: no wind}}$$

$$\boldsymbol{M}_{AB} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha \\ 0 & -\sin\alpha & \cos\alpha \end{pmatrix}$$

We must only rotate around the x-axis:

$$\left(\vec{\mathbf{F}}_A^G\right)_K = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & cos\,\mu_K & -sin\,\mu_K \\ 0 & sin\,\mu_K & cos\,\mu_K \end{pmatrix}}_{\boldsymbol{M}_{K\bar{K}}} \begin{pmatrix} -D \\ 0 \\ -L \end{pmatrix}_{\bar{K}} = \begin{pmatrix} -D \\ L\sin\mu_K \\ -L\cos\mu_K \end{pmatrix}_K$$

This single-axis rotation could in theory be part of an exam!

In this example, there is no (kinematic) bank angle: [(2) + II]

$$\left(\vec{\mathbf{F}}_A^G\right)_K = \left(\vec{F}_A^G\right)_A = \begin{pmatrix} -D \\ 0 \\ -L \end{pmatrix}_K$$

Force equilibrium z_K

Summarized:

$$\left(\overrightarrow{F_T}\right)_K = \left(\vec{F}_P^G\right)_K + \left(\vec{F}_A^A\right)_K + \left(\vec{F}_G^G\right)_K$$

$$\left(\vec{\mathbf{F}}_T^G\right)_K = \begin{pmatrix} X_T^G \\ Y_T^G \\ Z_T^G \end{pmatrix}_K = \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K + \begin{pmatrix} -D \\ 0 \\ -L \end{pmatrix}_K + \begin{pmatrix} -m\,g\sin\gamma_K \\ 0 \\ m\,g\cos\gamma_K \end{pmatrix}_K$$

We will use this expression in b)

a) What is the **required GPS velocity** such that **the z-component of the total force** in the **kinematic frame** is **zero**?

$$\left(F_{T,z}\right)_K = 0 \quad \rightarrow \quad V = ?$$

z-component must be zero: $0 - L + m\,g\cos\gamma_K = 0$

$$L = m\,g\cos\gamma_K$$

$$L = \bar{q}\,S_{ref}\,C_L = \frac{1}{2}\rho V_{TAS}^2 S_{ref}\,C_L$$

windfree: $V_{TAS} = \mathrm{V_K} = \mathrm{V_{GPS}}$

Force equilibrium z_K

$$L = m\,g\cos\gamma_K = \frac{1}{2}\rho V_{TAS}^2 S_{ref}\,C_L$$

$$V_{TAS}^2 = \frac{2\,mg}{S_{ref}\,C_L\,\boldsymbol{\rho}}\cos\gamma_K$$

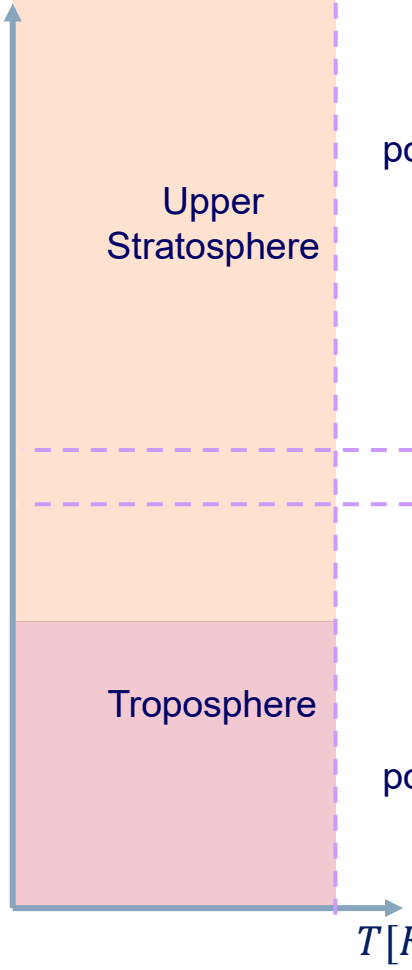
We must compute the density!

Geodetic Longitude	λ	11.669284 deg
Geodetic Latitude	μ	48.265906 deg
GPS-height	h	5000 m
Climb angle	$\boldsymbol{\gamma_K}$	**0, 5°**
Reference thrust MSL	T_i	5000 N
Reference density	ρ_i	$\rho_{S,MSL}$
Reference velocity	V_{ref}	80 kts
Density coefficient	n_ρ	0.75
Lift coefficient	$\boldsymbol{C_L}$	**1**
Drag coefficient	$C_{D,0}$	0,012
K - factor	k	0,016
Reference area	$\boldsymbol{S_{ref}}$	**20 m^2**
Aircraft mass	**m**	**2000 kg**

Gravitational acceleration:

$$g_S = 9.80665\,\frac{m}{s^2}$$

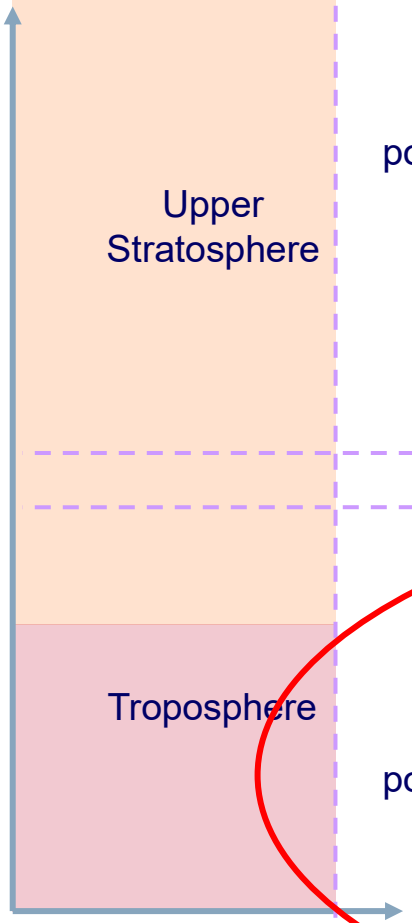
Key Facts ISA



Layer	Features Of Layer	Atmosphere Equations	Data	Height
Upper Stratosphere	polytrope	$\frac{T}{T_{20}} = \left[1 + \frac{\gamma_{str}}{T_{20}}(H_G - H_{G20})\right]$ $\frac{p}{p_{20}} = \left[1 + \frac{\gamma_{str}}{T_{20}}(H_G - H_{G20})\right]^{\frac{n_{str}}{n_{str}-1}}$ $\frac{\rho}{\rho_{20}} = \left[1 + \frac{\gamma_{str}}{T_{20}}(H_G - H_{GR})\right]^{\frac{1}{n_{str}-1}}$	$T_{20} = 216{,}65K$ $\rho_{20} = 0{,}088\frac{kg}{m^3}$ $p_{20} = 547{,}488Pa$ $H_{G20} = 20000m$ $n_{Str} = 0{,}972$ $\gamma_{Str} = +1{,}0e-3$	20 to 32 km
Troposphere	polytrope	$\frac{T}{T_{S,MSL}} = \left[1 + \frac{\gamma_{Tr}}{T_{S,MSL}}H_G\right]$ $\frac{p}{p_{S,MSL}} = \left[1 + \frac{\gamma_{Tr}}{T_{S,MSL}}H_G\right]^{\frac{n_{Tr}}{n_{Tr}-1}}$ $\frac{\rho}{\rho_{S,MSL}} = \left[1 + \frac{\gamma_{Tr}}{T_{S,MSL}}H_G\right]^{\frac{1}{n_{Tr}-1}}$	$T_{S,MSL} = 288{,}15K$ $\rho_{S,MSL} = 1{,}225\frac{kg}{m^3}$ $p_{S,MSL} = 101325Pa$ $H_{GS,MSL} = 0m$ $n_{Tr} = 1{,}235$ $\gamma_{Tr} = -6{,}5e-3$	-2 to 11 km

$g_S = 9{,}80665\frac{m}{s^2}$ $R = 287{,}05\frac{J}{kg\,K}$ $n = \frac{g_S}{g_S + \gamma R}$ $\gamma = -\left(\frac{n-1}{n}\right)\frac{g_S}{R}$

Key Facts ISA



Layer	Features Of Layer	Atmosphere Equations	Data	Height
Upper Stratosphere	polytrope	$\frac{T}{T_{20}} = \left[1 + \frac{\gamma_{str}}{T_{20}}(H_G - H_{G20})\right]$ $\frac{p}{p_{20}} = \left[1 + \frac{\gamma_{str}}{T_{20}}(H_G - H_{G20})\right]^{\frac{n_{str}}{n_{str}-1}}$ $\frac{\rho}{\rho_{20}} = \left[1 + \frac{\gamma_{str}}{T_{20}}(H_G - H_{G20})\right]^{\frac{1}{n_{str}-1}}$	$T_{20} = 216{,}65K$ $\rho_{20} = 0{,}088\frac{kg}{m^3}$ $p_{20} = 547{,}488Pa$ $H_{G20} = 20000m$ $n_{Str} = 0{,}972$ $\gamma_{Str} = +1{,}0e-3$	20 to 32 km
Troposphere	polytrope	$\frac{T}{T_{S,MSL}} = \left[1 + \frac{\gamma_{Tr}}{T_{S,MSL}}H_G\right]$ $\frac{p}{p_{S,MSL}} = \left[1 + \frac{\gamma_{Tr}}{T_{S,MSL}}H_G\right]^{\frac{n_{Tr}}{n_{Tr}-1}}$ $\frac{\rho}{\rho_{S,MSL}} = \left[1 + \frac{\gamma_{Tr}}{T_{S,MSL}}H_G\right]^{\frac{1}{n_{Tr}-1}}$	$T_{S,MSL} = 288{,}15K$ $\rho_{S,MSL} = 1{,}225\frac{kg}{m^3}$ $p_{S,MSL} = 101325Pa$ $H_{GS,MSL} = 0m$ $n_{Tr} = 1{,}235$ $\gamma_{Tr} = -6{,}5e-3$	-2 to 11 km

$g_S = 9{,}80665\frac{m}{s^2}$ $R = 287{,}05\frac{J}{kg\,K}$ $n = \frac{g_S}{g_S + \gamma R}$ $\gamma = -\left(\frac{n-1}{n}\right)\frac{g_S}{R}$

Troposphere: *R = MSL*

$$T_R = T_{S,MSL} = 288{,}15K$$

$$\rho_R = \rho_{S,MSL} = 1{,}225\,\frac{kg}{m^3}$$

$$p_R = p_{S,MSL} = 101325Pa$$

$$H_{GR} = H_{GS,MSL} = 0$$

Density:

$$\frac{\rho(H_G)}{\rho_R} = \left[1 + \frac{\gamma}{T_R}(H_G - H_{GR})\right]^{\frac{1}{n-1}}$$

$$\rho(H_G) = \rho_R \left[1 + \frac{\boldsymbol{\gamma}}{T_R} H_G\right]^{\frac{1}{n-1}}$$

$$\gamma = -\left(\frac{n-1}{n}\right)\frac{g_S}{R} = -6{,}5e-3$$

Specific gas constant: $R = 287.05\,\frac{J}{kg\,K}$

Gravitational acceleration: $g_s = 9.80665\ \mathrm{m/s^2}$

Polytropic exponent (Troposphere): $n = 1.235$

troposphere: *R = MSL*

$T_R = T_{S,MSL} = 288{,}15K$

$\rho_R = \rho_{S,MSL} = 1{,}225\,\frac{kg}{m^3}$

$p_R = p_{S,MSL} = 101325Pa$

$H_{GR} = H_{GS,MSL} = 0$

$\gamma = -6{,}5e-3$

Polytropic exponent (Troposphere): $n = 1.235$

Specific gas constant: $R = 287.05\,\frac{J}{kg\,K}$

Gravitational acceleration: $g_s = 9.80665\ \mathrm{m/s^2}$

Density:

$$\rho(H_G) = \rho_R \left[1 + \frac{\gamma}{T_R} H_G\right]^{\frac{1}{n-1}} = 0{,}736\,\frac{kg}{m^3}$$

geopotential height:

$$H_G = \frac{r_E \cdot h}{r_E + h} = 4996\,m$$

$r_E = 6356766\,m$

$h = 5000\,m$

Force equilibrium z_K

$$V_{TAS}^2 = \frac{2\,mg}{S_{ref}\,C_L\,\boldsymbol{\rho}}\cos\gamma_K$$

We must compute the density! $\rho = 0{,}736\ \frac{kg}{m^3}$

Now we know everything

$$V_{TAS} = \sqrt{\frac{2\,m\,g\cos\gamma_K}{\rho\,S_{ref}\,C_L}} = \sqrt{\frac{2\cdot\ 2000\cdot\ 9.80665\cdot\cos\left(0.5\cdot\frac{pi}{180}\right)}{0{,}736\cdot\ 20\cdot\ 1}} = 51{,}62\frac{m}{s}$$

Geodetic Longitude	λ	11.669284 deg
Geodetic Latitude	μ	48.265906 deg
GPS-height	h	5000 m
Climb angle	$\boldsymbol{\gamma_K}$	**0, 5°**
Reference thrust MSL	T_i	5000 N
Reference density	ρ_i	$\rho_{s,MSL}$
Reference velocity	V_{ref}	80 kts
Density coefficient	n_ρ	0.75
Lift coefficient	$\boldsymbol{C_L}$	**1**
Drag coefficient	$C_{D,0}$	0,012
K - factor	k	0,016
Reference area	$\boldsymbol{S_{ref}}$	**20 m^2**
Aircraft mass	**m**	**2000 kg**

windfree: $V_{TAS} = V_K = V_{GPS}$

Gravitational acceleration:

$$g_S = 9.80665\frac{m}{s^2}$$

Exercise 1: Pointmass-Equations and Forces

A 1-engine propeller aircraft (piston engine) is flying in a windfree standard ISA-atmosphere without a bank angle in a straight flight with a climb angle γ_K of 0.5°. You are given the following data.

In the following you can assume that no aerodynamic side-forces act on the aircraft and that the kinematic bank angle μ_K equals the euler bank angle ϕ.

a) What is the required GPS velocity such that the z-component of the total force in the kinematic frame is zero?

b) Calculate the total force acting on the aircraft. State the result in a suitable coordinate frame.

c) Calculate the current time derivative of the aircraft states.

d) What happens to the time derivatives if the aircraft flies with a kinematic bank angle $\mu_K = 30°$ if all other states remain the same?

Total force

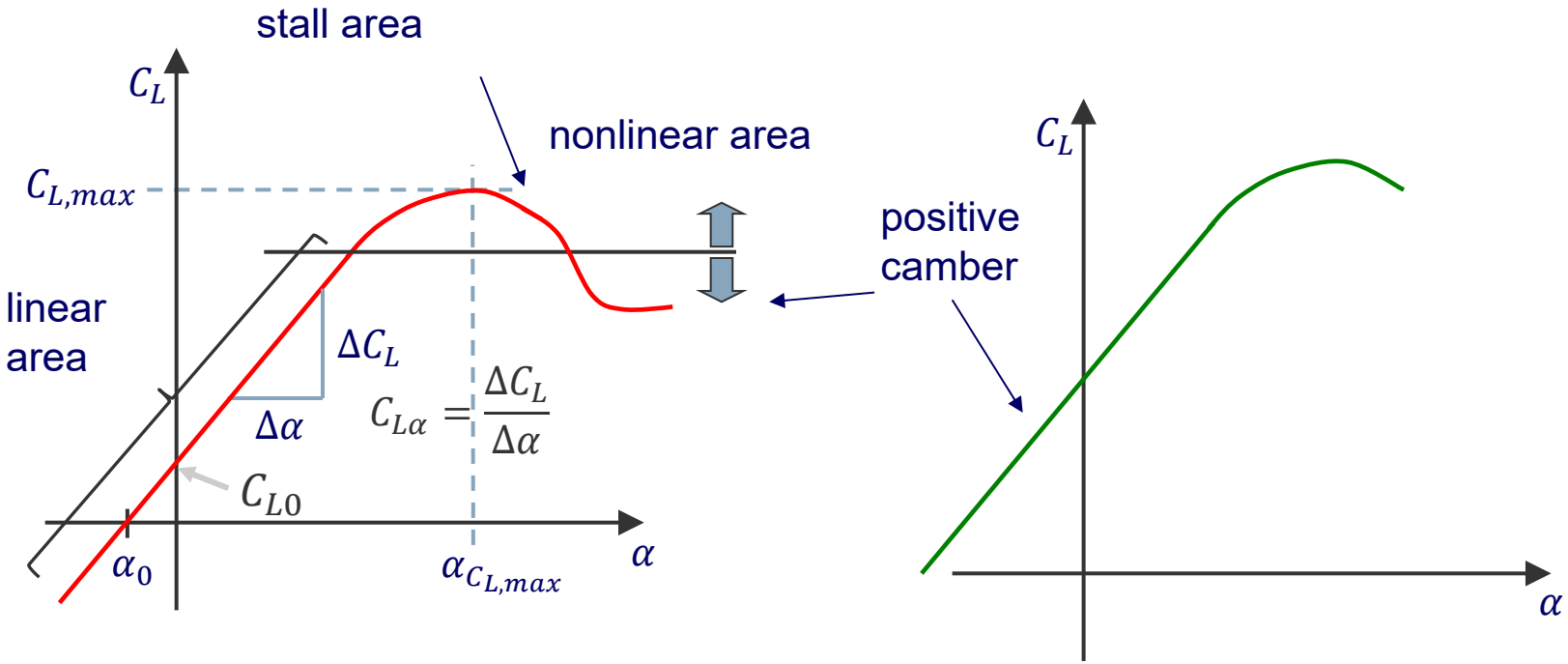
b) Calculate the total force acting on the aircraft. Give the force in a suitable coordinate system.

Note:
K-frame is suitable for point-mass equations of motion

From a):

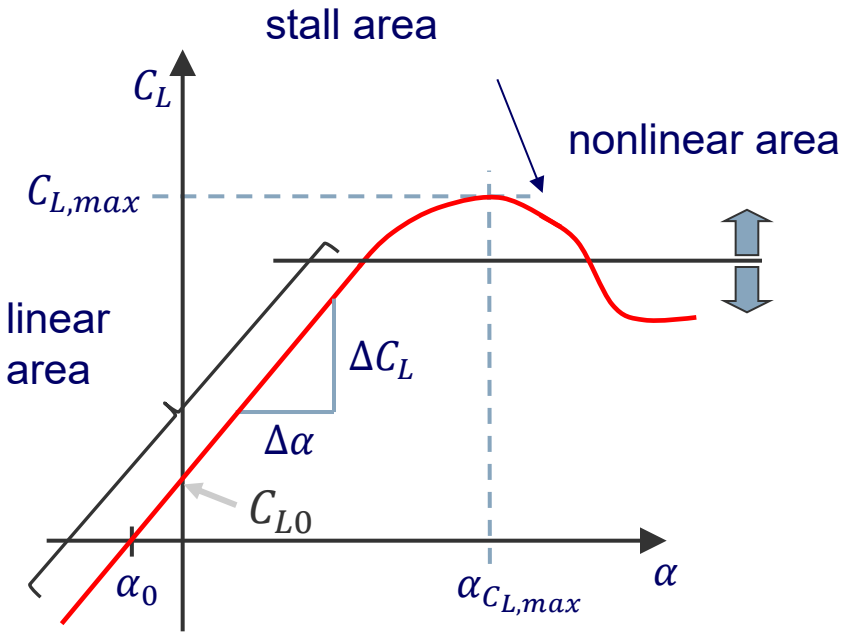
$$\left(\vec{\mathbf{F}}_T^G\right)_K = \begin{pmatrix} X_T^G \\ Y_T^G \\ Z_T^G \end{pmatrix}_K = \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K + \begin{pmatrix} -D \\ 0 \\ -L \end{pmatrix}_K + \begin{pmatrix} -m\, g \sin\gamma_K \\ 0 \\ m\, g \cos\gamma_K \end{pmatrix}_K$$

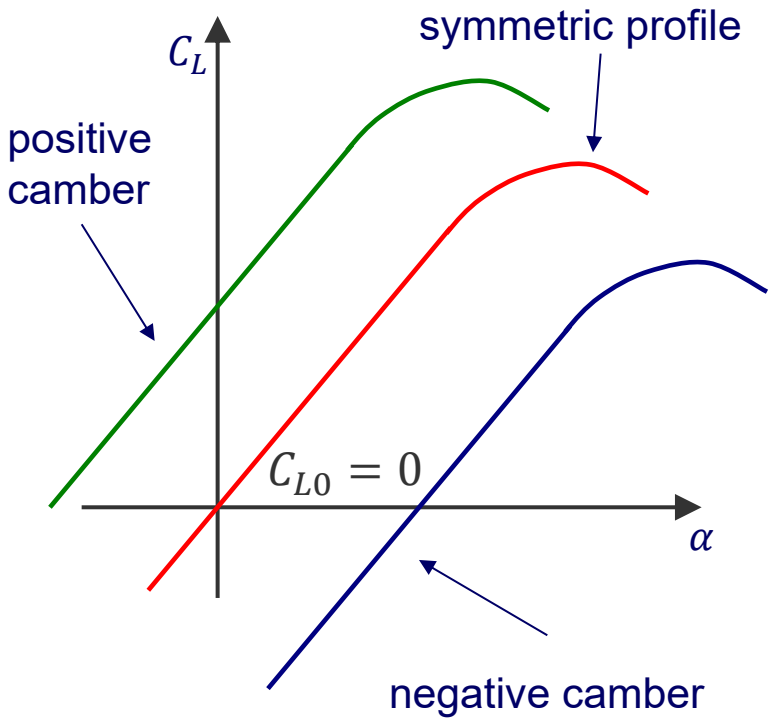
C_L-α – Diagram



a) **Note down the function $C_L(\alpha)$ in the linear region.**
b) **Draw the $C_L(\alpha)$ diagrams qualitatively for a negative chamber and a symmetric chamber**

C_L-α – Diagram





$$C_L \approx C_{L0} + \frac{\Delta C_L}{\Delta \alpha}\,\alpha$$

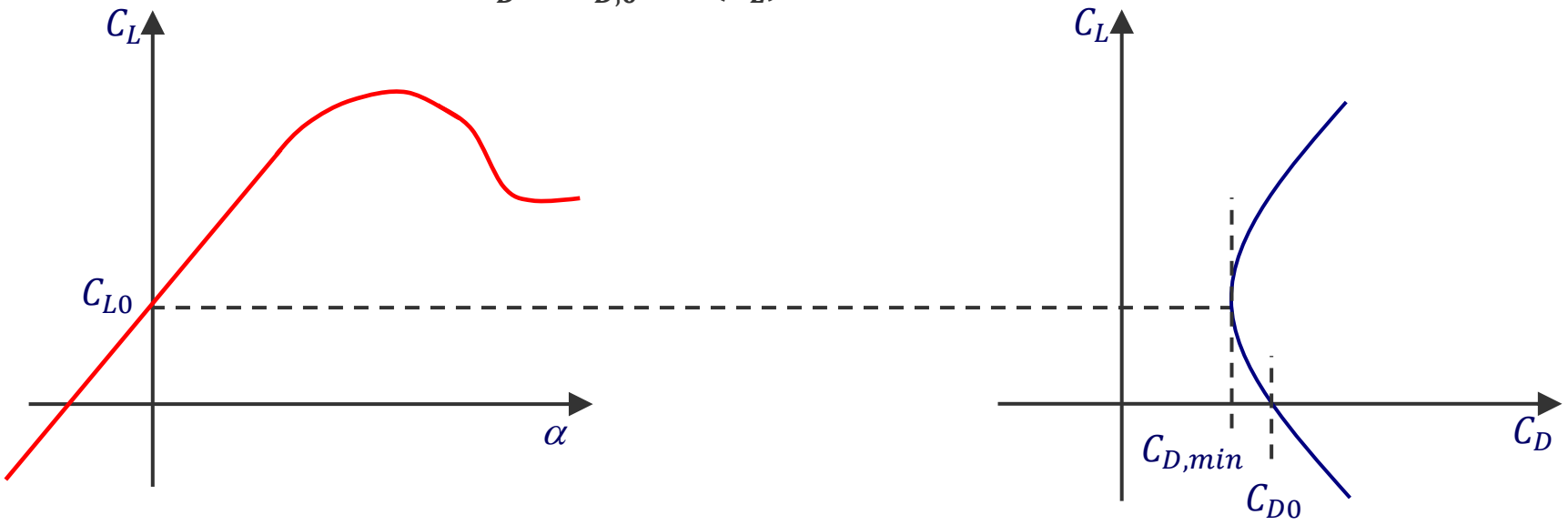
$$C_L \approx C_{L0} + C_{L\alpha} \cdot \alpha$$

C_{L0}:Zero lift coefficient
$C_{L\alpha}$: lift deviation over α

Lift and Drag coefficients

Lift induced drag:

$$C_D = C_{D,0} + k(C_L)^2$$



quadratic polar:

$$C_D = C_{D,0} + k(C_L - C_{L0})^2$$

$C_{L0} = 0$ for symmetric profiles, see previous slide

symmetric quadratic polar:

$$C_D = C_{D,0} + k(C_L)^2$$

Engine Types – Mathematical modelling

Behavior of an engine is described by

$$\frac{T}{T_i} = \left(\frac{V}{V_i}\right)^{n_V} \left(\frac{\rho}{\rho_i}\right)^{n_\rho}$$

Thrust T, Velocity V

Dependency of thrust on speed:

$n_V = -1$: classic reciprocrating engine with propeller	$n_V = -0.75$: Turboprop
$n_V = 0$: Turbojet engine (subsonic)	$n_V = -0.25$: Turbofan

Dependency of thrust on height:

$n_\rho = 0{,}7 - 0{,}8$:	for air breathing engines at heights of $0 < H < 11\ km$
$n_\rho = 1{,}0$:	for air breathing engines at heights of $11\ km < H < 20\ km$

b) Calculate the total force acting on the aircraft. Give the force in a suitable coordinate system.

Note:
K-frame is suitable for point-mass equations of motion

From a): $$\left(\vec{\mathbf{F}}_T^G\right)_K = \begin{pmatrix} X_T^G \\ Y_T^G \\ Z_T^G \end{pmatrix}_K = \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K + \begin{pmatrix} -D \\ 0 \\ -L \end{pmatrix}_K + \begin{pmatrix} -m\, g \sin\gamma_K \\ 0 \\ m\, g \cos\gamma_K \end{pmatrix}_K$$

We use known relations:

$$\frac{T}{T_i} = \left(\frac{V}{V_i}\right)^{n_V} \left(\frac{\rho}{\rho_i}\right)^{n_\rho}$$

$$\mathrm{D} = \bar{q}\, S_{ref}\, C_D = \frac{1}{2}\rho V_{TAS}^2 S_{ref}\, C_D$$

$$C_D = C_{D,0} + k(C_L)^2$$

Note: Know these relations!

Note: Know which n_V is used for which type of aircraft!

Here: $n_V = -1$
1-engine propeller aircraft (**piston engine**)

Total force

b) Calculate the total force acting on the aircraft. Give the force in a suitable coordinate system.

$$\left(\vec{\mathbf{F}}_T^G\right)_K = \begin{pmatrix} X_T^G \\ Y_T^G \\ Z_T^G \end{pmatrix}_K = \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K + \begin{pmatrix} -D \\ 0 \\ -L \end{pmatrix}_K + \begin{pmatrix} -m\,g\sin\gamma_K \\ 0 \\ m\,g\cos\gamma_K \end{pmatrix}_K = \begin{pmatrix} ? \\ 0 \\ 0 \end{pmatrix}_K$$

- From a) : z-component must be zero
- Task: no sideforce

For the **drag**:

$$C_D = C_{D,0} + k(C_L)^2$$

$$D = \frac{1}{2}\rho V^2 S_{ref}\, C_D$$

Force equilibrium z_K

$$C_D = C_{D,0} + k(C_L)^2 = 0.012 + 0.016 \cdot (1)^2 = 0.028$$

$$D = \frac{1}{2}\rho V^2 S_{ref}\, C_D = \left(\frac{1}{2} \cdot 0.736 \cdot 51{,}62^2 \cdot 20 \cdot 0.028\right) N = 549{,}15\ N$$

from a)

Geodetic Longitude	λ	11.669284 deg
Geodetic Latitude	μ	48.265906 deg
GPS-height	$h = h_{geod}$	5000 m
Climb angle	γ_K	0,5°
Reference thrust MSL	T_i	5000 N
Reference density	ρ_i	$\rho_{s,MSL}$
Reference velocity	V_{ref}	80 kts
Density coefficient	n_ρ	0.75
Lift coefficient	$\boldsymbol{C_L}$	**1**
Drag coefficient	$\boldsymbol{C_{D,0}}$	**0,012**
K - factor	$\boldsymbol{k}$	**0,016**
Reference area	S_{ref}	20 m^2
Aircraft mass	m	2000 kg

Total force

b) Calculate the total force acting on the aircraft. Give the force in a suitable coordinate system.

$$\left(\vec{\mathbf{F}}_T^G\right)_K = \begin{pmatrix} X_T^G \\ Y_T^G \\ Z_T^G \end{pmatrix}_K = \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K + \begin{pmatrix} -D \\ 0 \\ -L \end{pmatrix}_K + \begin{pmatrix} -m\,g\sin\gamma_K \\ 0 \\ m\,g\cos\gamma_K \end{pmatrix}_K = \begin{pmatrix} ? \\ 0 \\ 0 \end{pmatrix}_K$$

- From a) : z-component must be zero
- Task: no sideforce

For the **thrust**:

$$\frac{T}{T_i} = \left(\frac{V}{V_i}\right)^{n_V} \left(\frac{\rho}{\rho_i}\right)^{n_\rho}$$

Here, piston engine: → $n_V = -1$

Force equilibrium z_K

$$\frac{T}{T_i} = \left(\frac{V}{V_i}\right)^{n_V} \left(\frac{\rho}{\rho_i}\right)^{n_\rho}$$

Here, piston engine: → $n_V = -1$

from a)

$$T = 5000 \left(\frac{51{,}62}{80 \cdot 0{,}5144}\right)^{-1} \left(\frac{0{,}736}{1{,}225}\right)^{0{,}75} N = 2720{,}19\ N$$

Geodetic Longitude	λ	11.669284 deg
Geodetic Latitude	μ	48.265906 deg
GPS-height	$h = h_{geod}$	5000 m
Climb angle	γ_K	0,5°
Reference thrust MSL	$\boldsymbol{T_i}$	**5000 N**
Reference density	$\boldsymbol{\rho_i}$	$\boldsymbol{\rho_{s,MSL}}$
Reference velocity	$\mathbf{V_i}$	**80 kts**
Density coefficient	$\mathbf{n_\rho}$	**0.75**
Lift coefficient	C_L	1
Drag coefficient	$C_{D,0}$	0,012
K - factor	k	0,016
Reference area	S_{ref}	20 m^2
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Total force

b) Calculate the total force acting on the aircraft. Give the force in a suitable coordinate system.

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- From a) : z-component must be zero
- Task: no sideforce

For the **gravitational force**:

$$-m\ g \sin\gamma_K$$

Force equilibrium z_K

$$-m\ g \sin \gamma_K = -171{,}16\text{N}$$

Geodetic Longitude	λ	11.669284 deg
Geodetic Latitude	μ	48.265906 deg
GPS-height	$h = h_{geod}$	5000 m
Climb angle	γ_K	**0, 5°**
Reference thrust MSL	T_i	5000 N
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Drag coefficient	$C_{D,0}$	0,012
K - factor	k	0,016
Reference area	S_{ref}	20 m^2
Aircraft mass	m	**2000 kg**

Gravitational acceleration:

$$g_S = 9.80665 \frac{m}{s^2}$$

Total force

b) Calculate the total force acting on the aircraft. Give the force in a suitable coordinate system.

$$\left(\vec{\mathbf{F}}_T^G\right)_K = \begin{pmatrix} X_T^G \\ Y_T^G \\ Z_T^G \end{pmatrix}_K = \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K + \begin{pmatrix} -D \\ 0 \\ -L \end{pmatrix}_K + \begin{pmatrix} -m\ g \sin\gamma_K \\ 0 \\ m\ g \cos\gamma_K \end{pmatrix}_K = \begin{pmatrix} 1999{,}88\ N \\ 0 \\ 0 \end{pmatrix}_K$$

Exercise 1: Pointmass-Equations and Forces

A 1-engine propeller aircraft (piston engine) is flying in a windfree standard ISA-atmosphere without a bank angle in a straight flight with a climb angle γ_K of 0.5°. You are given the following data.

In the following you can assume that no aerodynamic side-forces act on the aircraft and that the kinematic bank angle μ_K equals the euler bank angle ϕ.

a) What is the required GPS velocity such that the z-component of the total force in the kinematic frame is zero?

b) Calculate the total force acting on the aircraft. State the result in a suitable coordinate frame.

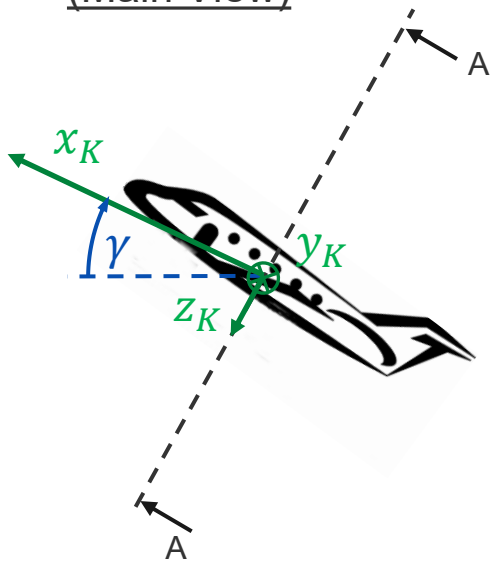
c) Calculate the current time derivative of the aircraft states.

d) What happens to the time derivatives if the aircraft flies with a kinematic bank angle $\mu_K = 30°$ if all other states remain the same?

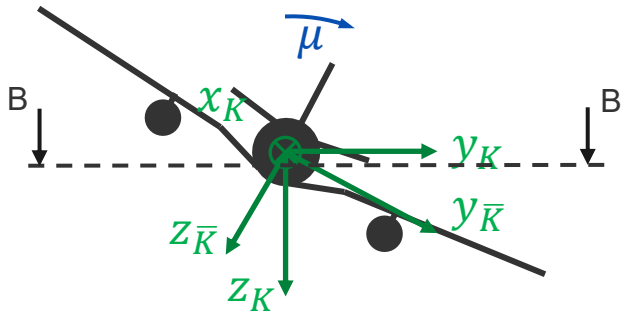
Stationary Balance of Forces

All forces are marked on the aircraft:
In the first step, dynamic changes are not considered yet:

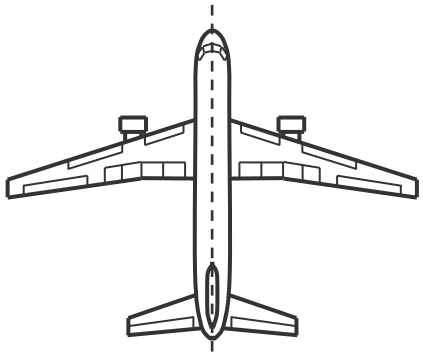
Side View
(Main View)



View A-A



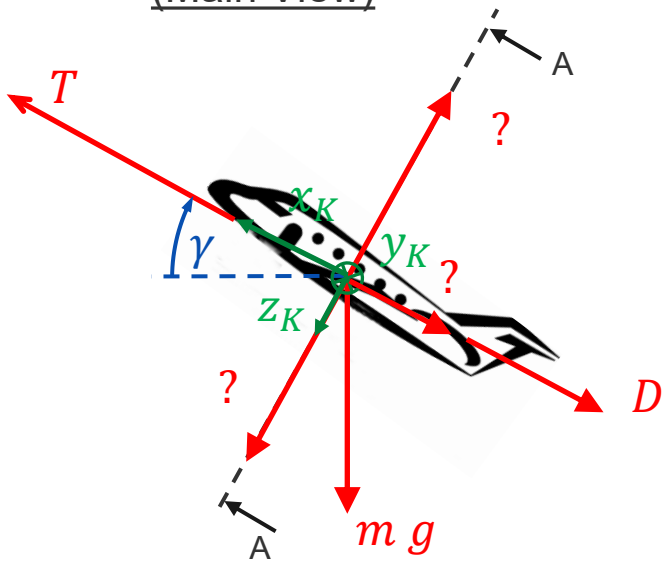
View B-B



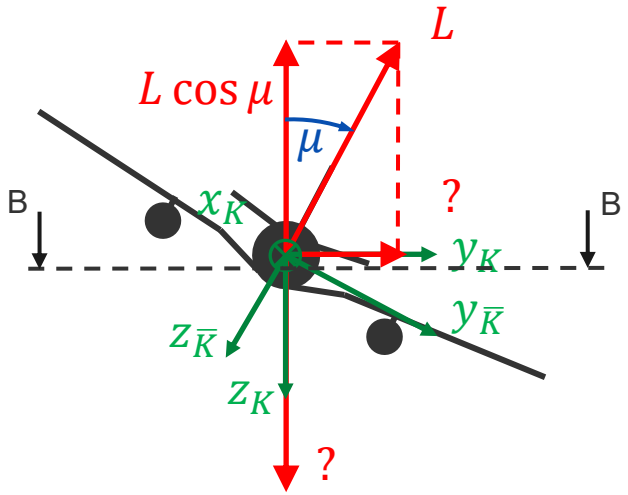
Stationary Balance of Forces

All forces are marked on the aircraft:
In the first step, dynamic changes are not considered yet:

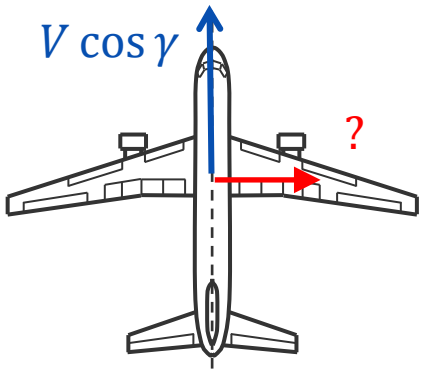
Side View
(Main View)



View A-A



View B-B

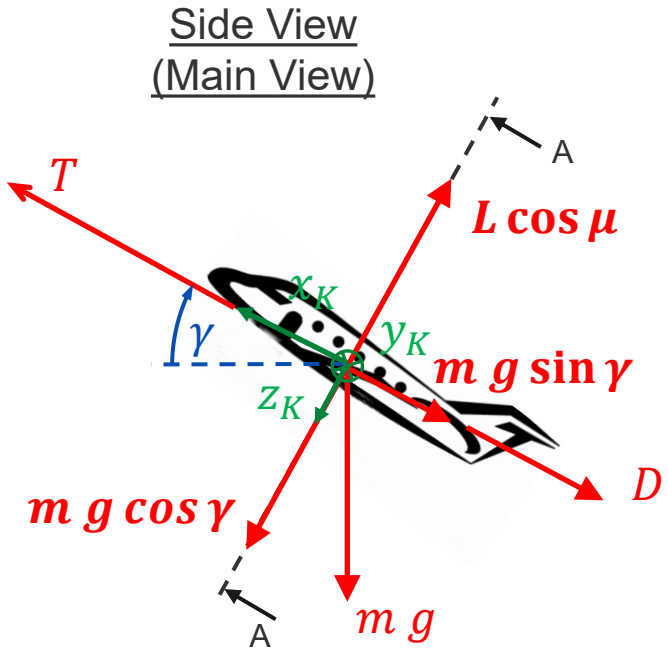




Can you note down the missing expressions? (question marks)

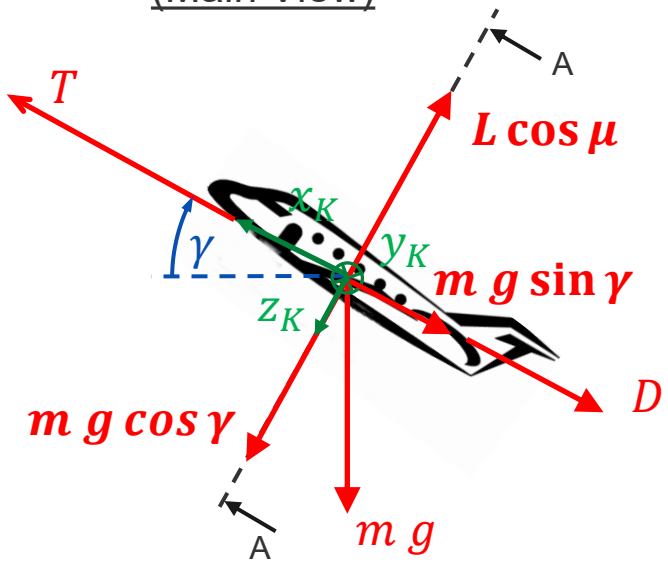
Stationary Balance of Forces

All forces are marked on the aircraft:
In the first step, dynamic changes are not considered yet:

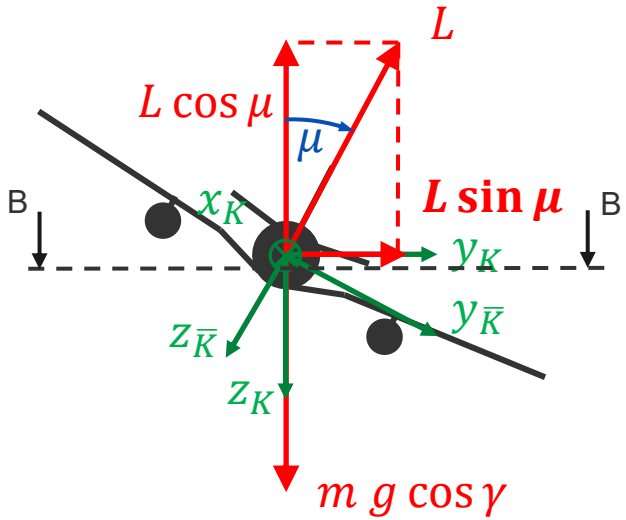


All forces are marked on the aircraft:
In the first step, dynamic changes are not considered yet:

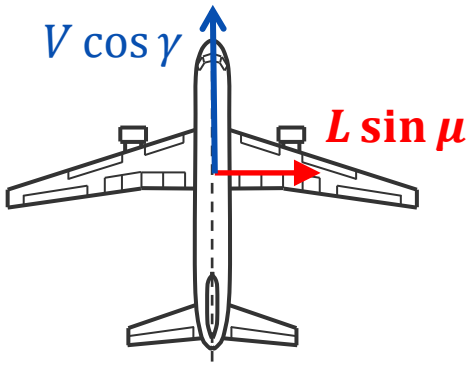
Side View
(Main View)



View A-A



View B-B

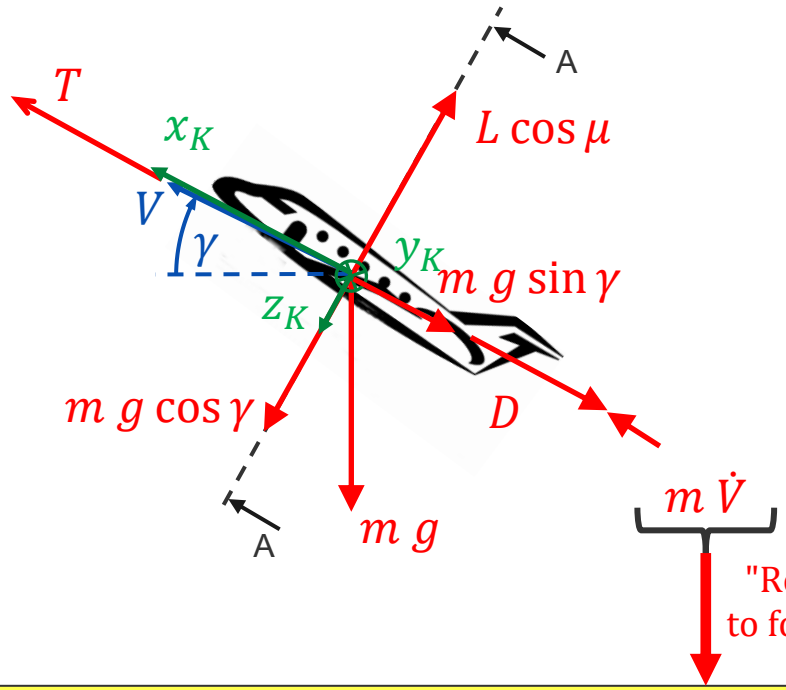


Dynamic Balance of Forces

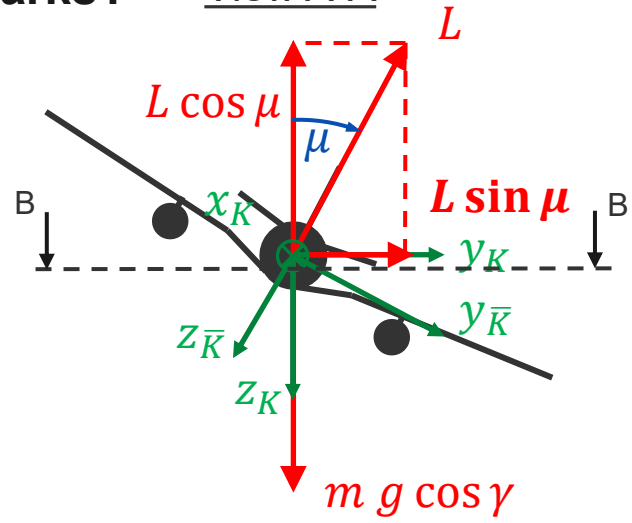
For simplification flight path changes (turn and flight path angle change) are considered in the next step.

$$\left(\vec{\mathbf{F}}_T^G\right)_K = \begin{pmatrix} X_T^G \\ Y_T^G \\ Z_T^G \end{pmatrix}_K = \begin{pmatrix} m\dot{V} \\ ? \\ ? \end{pmatrix}_K$$

$$= \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K + \begin{pmatrix} -D \\ ? \\ ? \end{pmatrix}_K + \begin{pmatrix} -m\,g\sin\gamma_K \\ 0 \\ ? \end{pmatrix}_K$$



Can you fill the red question marks? View A-A



The change of linear momentum only shows x-contributions when considering derivatives with respect to the K-frame

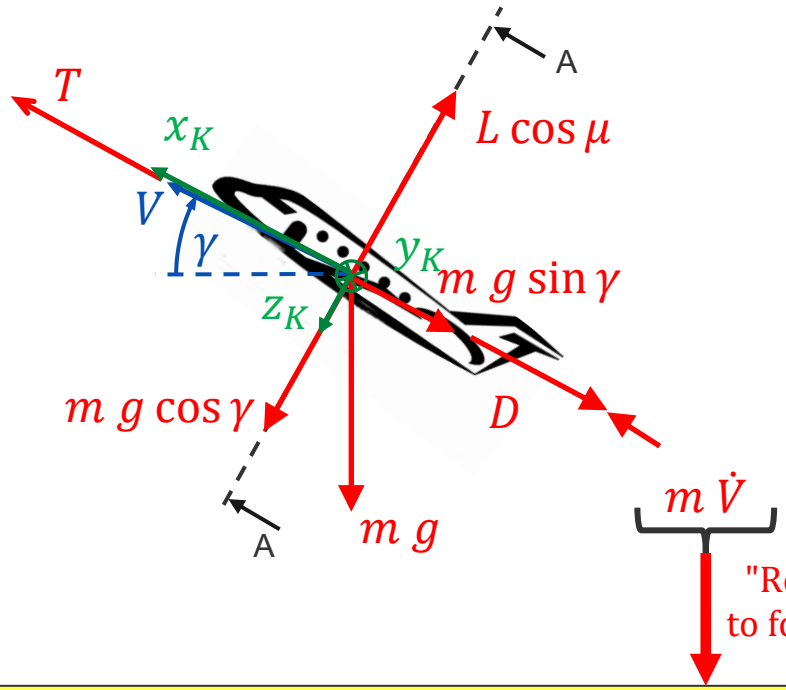
$$\left(\dot{\vec{V}}^G\right)_K = \begin{bmatrix} \dot{V} \\ 0 \\ 0 \end{bmatrix}_K$$

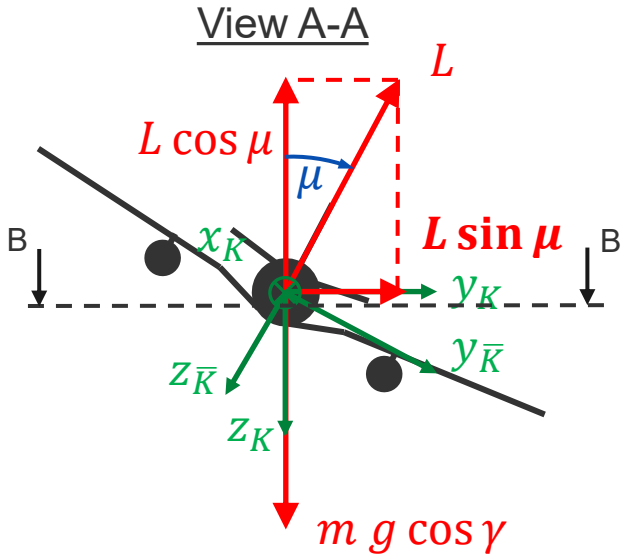
Dynamic Balance of Forces

For simplification flight path changes (turn and flight path angle change) are considered in the next step.

$$\left(\vec{\mathbf{F}}_T^G\right)_K = \begin{pmatrix} X_T^G \\ Y_T^G \\ Z_T^G \end{pmatrix}_K = \begin{pmatrix} m\dot{V} \\ ? \\ ? \end{pmatrix}$$

$$= \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K + \begin{pmatrix} -D \\ L\sin\mu \\ -L\cos\mu \end{pmatrix}_K + \begin{pmatrix} -m\,g\sin\gamma_K \\ 0 \\ m\,g\cos\gamma_K \end{pmatrix}_K$$

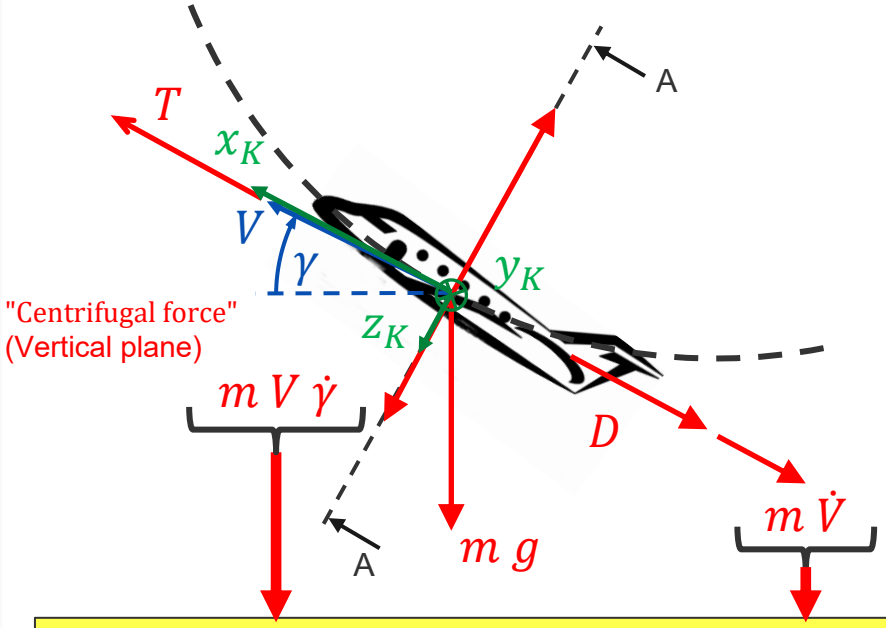


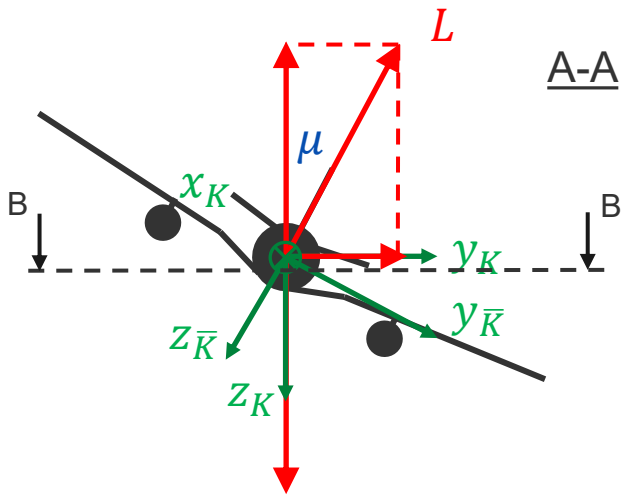


The change of linear momentum only shows x-contributions when considering derivatives with respect to the K-frame

$$\left(\dot{\vec{V}}^G\right)_K = \begin{bmatrix} \dot{V} \\ 0 \\ 0 \end{bmatrix}_K$$

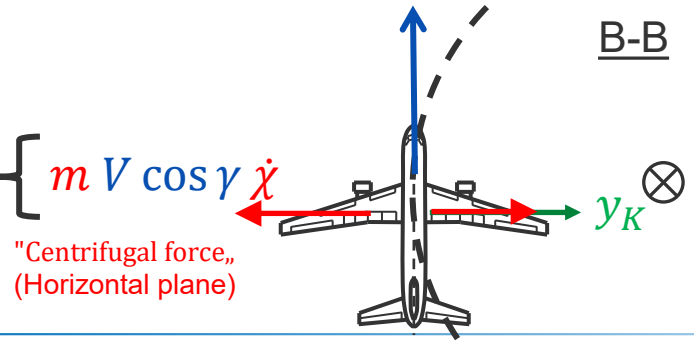
For defining the full motion of the aircraft, time derivative of course and flight path angle have to be taken into account.





Accellerations expressed in kinematic system K:

$$\left(\vec{V}^G\right)_K = \begin{bmatrix} \dot{V} \\ 0 \\ 0 \end{bmatrix}_K + \begin{bmatrix} 0 \\ V\cos\gamma\,\dot{\chi} \\ -V\cdot\dot{\gamma} \end{bmatrix}_K$$



Equation of Motion Translation

$$\left(\vec{\mathbf{F}}_T^G\right)_K = \begin{pmatrix} X_T^G \\ Y_T^G \\ Z_T^G \end{pmatrix}_K = \begin{pmatrix} m\dot{V} \\ mV\cos\gamma\,\dot{\chi} \\ -mV\cdot\dot{\gamma} \end{pmatrix}_K = \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K + \begin{pmatrix} -D \\ L\sin\mu \\ -L\,cos\mu \end{pmatrix}_K + \begin{pmatrix} -m\,g\sin\gamma_K \\ 0 \\ m\,g\cos\gamma_K \end{pmatrix}_K$$

Solving above equations for the quantities $\dot{V}$, $\dot{\chi}$ und $\dot{\gamma}$ leads to the equations of motion.

Note: You have to know these!

$$\Longrightarrow \dot{V} = \frac{T-D}{m} - g\sin\gamma$$

$$\dot{\chi} = \frac{L\sin\mu}{m\,V\cos\gamma}$$

$$\dot{\gamma} = \frac{L\cos\mu}{m\,V} - \frac{g}{V}\cos\gamma$$

$$L = \bar{q}\,S\,C_L$$

$$\mathrm{D} = \bar{q}\,S\,C_D$$

$$\bar{q} = \frac{1}{2}\rho\,V^2$$

$$C_D = C_{D_o} + k\,C_L^2$$

$$T = L_P\,V^{n_V}$$

Note: Know which n_V is used for which type of aircraft!

c) Calculate the current time derivative of the aircraft states:

From lecture:

$$\dot{V} = \underbrace{\frac{T - D}{m} - g \sin\gamma}_{X_T^G / m}$$

Velocity

$$\dot{\chi} = \frac{L \sin\mu}{m\, V \cos\gamma}$$

Course angle

Course is changed by a change in lift (i.e. a force perpendicular to the current flight direction)

$$\dot{\gamma} = \frac{L \cos\mu}{m\, V} - \frac{g}{V}\cos\gamma$$

Climb angle

Same story, but around another direction.

c) Calculate the current time derivative of the aircraft states:

see b)

Task: $\mu_K = 0$

$$L = \frac{1}{2}\rho V_{TAS}^2 S_{ref}\, C_L = mg\cos\gamma_k$$

see a)

$$\dot{V} = \frac{T - D}{m} - g\sin(\gamma_K) = 0{,}9999\,\frac{m}{s^2}$$

$$\dot{\chi} = \frac{L\sin\mu_K}{m\,V\cos(\gamma_K)} = 0\,\frac{rad}{s}$$

$$\dot{\gamma} = \frac{L\cos\mu_K}{m\,V} - \frac{g}{V}\cos(\gamma_K) = 0\,\frac{rad}{s}$$

"Qualitatively, the motion can be described as an **accelerated climb**. Neither the angles χ nor γ change at this moment, so there is no path curvature. The velocity V changes, so it is an accelerated motion. "

CAUTION: We have only taken a snapshot of the current aircraft situation. This does not mean that the entire flight is necessarily a straight climb. For example, the climb **changes the altitude**, but thereby also the **density**, and we would get a **changed lift force** in the next instant, which then also **changes the entire force balance**. The consideration of actually stationary flight conditions follows in chapter 6 of the lecture.

Exercise 1: Pointmass-Equations and Forces

A 1-engine propeller aircraft (piston engine) is flying in a windfree standard ISA-atmosphere without a bank angle in a straight flight with a climb angle γ_K of 0.5°. You are given the following data.

In the following you can assume that no aerodynamic side-forces act on the aircraft and that the kinematic bank angle μ_K equals the euler bank angle ϕ.

a) What is the required GPS velocity such that the z-component of the total force in the kinematic frame is zero?
b) Calculate the total force acting on the aircraft. State the result in a suitable coordinate frame.
c) Calculate the current time derivative of the aircraft states.
d) What happens to the time derivatives if the aircraft flies with a kinematic bank angle $\mu_K = 30°$ if all other states remain the same?

d) What happens to the time derivatives when the aircraft flies with a **kinematic bank angle of 30°** instead of 0°, if all other aircraft states and forces remain the same?

$$\dot{V} = \frac{T - D}{m} - g \sin\left(0{,}5 \cdot \frac{\pi}{180}\right) = 0{,}9999 \frac{m}{s^2}$$

„Nothing changes here!“

$$\dot{\chi} = \frac{L \sin\left(30 \cdot \frac{\pi}{180}\right)}{m\, V \cos(0{,}5 \cdot \frac{\pi}{180})} = 0{,}095 \frac{rad}{s}$$

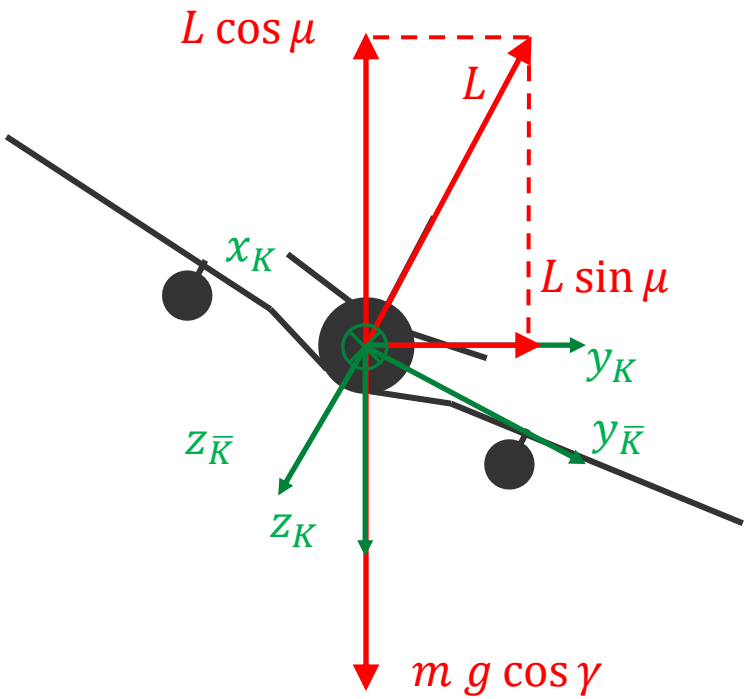
„We are in a turn!“

$$\dot{\gamma} = \frac{L \cos\left(30 \cdot \frac{\pi}{180}\right)}{m\, V} - \frac{g}{V} \cos\left(0{,}5 \cdot \frac{\pi}{180}\right) = -0{,}0255 \frac{rad}{s}$$

„We cannot keep the climb angle anymore!“

The reason is that the lift vector can now no longer be used completely to compensate for the weight force due to the curved flight. Instead, a part of the lift is applied to the path curvature. Therefore, the aircraft slowly reduces its angle of climb.

State time derivatives turn



- *Angle of attack remains unchanged*
- *Magnitude of L remains unchanged*
- *Direction of L changes*
- *Entering a turn* $\dot{\chi}_K$
- *Losing climb angle* $\dot{\gamma}_K$



Introduction To Flight Mechanics

Tutorial 4